

- High loud sounds cause:

- a. ↑↑ frequency of firing (A.P.) in auditory nerve fibers
- b. Stimulation of larger portion of the basilar membrane

3. Locality discrimination:

- a. Intensity difference: The sound is louder on the ear closer to its source
- b. Phase difference: At low frequencies, there is phase (time) difference in the sound waves reaching the 2 ears
- c. The ear pinna helps in discrimination of the sound source

4. Pattern discrimination:

Auditory cortex can recognize a sequence of tones following each other in a particular pattern, *e.g.*, music

Lecture (22): Chemical Senses (Smell & Taste)

(Dr. Wagdy Nashaat Habeeb)

- Generally classified as visceral senses due to their close association with GI functions
- Physiologically, they are related to each other:
 - (1) Flavors of foods are a combination of their taste & smell
 - (2) Both of taste & smell receptors are chemoreceptors which are stimulated by molecules
- Anatomically, they are quite different:
 - (1) Smell receptors are distant receptors (teleceptors), its pathway has no relay

in the thalamus & there is no neocortical projection area for olfaction

(2) Taste pathway relays in the thalamus & then project to the postcentral gyrus along with those for touch & pressure sensations from the mouth

(A) Smell (Olfaction)

➤ Importance of smell:

1. It determines the flavor of food
 2. It is essential for GI functions
 3. It helps to avoid dangerous liquids & volatile substances
 4. It plays a role in sexual behavior (in animals & may be in humans)
- N.B.* Smell sense is more acute in ♀ especially at time of ovulation
5. It triggers long-term memories

➤ Olfactory receptors:

*Site: in olfactory mucosa in sup. nasal concha adjacent to nasal septum

*Properties:

1. Chemoreceptors
2. Sensitivity: ↑↑
3. Discrimination of odors: ↑↑ (up to 10.000 different odors)
4. Discrimination of intensity: ↓↓ (conc. should be changed by $\geq 30\%$ to detect the difference)
5. Specificity: no (1 receptor can respond to many odors)
6. Adaptation: very rapid

*Odorant substance: It must be:

1. Volatile: to reach the olfactory area
2. Water soluble: to pass through the mucus layer
3. Lipid soluble: as olfactory cilia are composed of lipids

*Mechanism of stimulation:

Odorant substance → comes in contact with olfactory area → dissolves into mucus layer → binds with olfactory cilia → (+) adenylyl cyclase → ↑↑ cAMP → opens ligand-gated Na⁺ channels → ↑↑ Na⁺ inflow → depolarization of olfactory receptors (= receptor potential)

➤ Sniffing: (Semi-reflex)

*Stimulus: When new odor attracts attention

*Response: Contraction of the lower parts of the nares on the nasal septum

*Significance: To help the deflection of air stream upwards

(Normally, the region of olfactory receptors is poorly ventilated as air reaches it through convection eddy currents)

➤ **Test for smell:** By familiar non-irritant substance (to avoid stimulation of 5th CN)

➤ Abnormalities of smell:

1. Anosmia: = absence of smell sense (temporary or permanent)

*Causes:

(a) Congenital

(b) Acquired: e.g., with COVID-19, URTI (as common cold & sinusitis), continuous use of VC nasal sprays, nasal polyps, head trauma with damage of ethmoid bone & radiation therapy to the head and neck

2. Hyposmia: = ↓↓ smell acuity

*Causes: Allergies, nasal polyps & viral infections

3. Hypersomnia: = ↑↑ smell acuity

*Causes: Pregnancy & autoimmune diseases (as with Addison's disease & SLE)

4. Dysosmia: = distorted smell sense, includes:

i. Parosmia: = distorted smell sense in the presence of an odor

*Causes: URTI

ii. Phantosmia: = distorted smell sense in the absence of an odor (*i.e.*, olfactory hallucinations)

*Causes: Migraine & psychiatric diseases (as schizophrenia)

iii. Cacosmia: = continuous unpleasant smell sensation

*Causes: Sinusitis

(B) Taste (Gustation)

➤ **Taste receptors:** "Taste buds"

*Site:

1. Mainly: in tongue papillae on the dorsum of tongue

2. Few: in palate, epiglottis & pharynx

*Properties:

1. Each taste bud may contain > 1 type of receptors (with predominance of 1 type)

2. Discrimination of intensity: ↓↓ (conc. should be changed by ≥ 30 % to detect the difference)

3. Specificity:

(a) At low conc., each taste bud responds to 1 primary stimulus

(b) At high conc., most taste buds respond to > 1 stimulus

4. Adaptation: rapid

*Mechanism of stimulation:

Tasteful substances must be dissolved in saliva → reach microvilli of receptors *via* pores on top of taste buds → receptor potential (usually by depolarization) → generates AP in sensory nerve fibers

*Basic taste modalities:

Basic taste modality	<i>e.g.,</i>	Site	Mechanism of stimulation of taste receptors	
1. <u>Sweet:</u>	Sugars & glycerol	Tip of tongue	Depolarization of taste receptors by:	Binding to membrane receptor → (+) adenylyl cyclase → ↑↑ cAMP → closes K ⁺ channel → ↓↓ K ⁺ outflow
2. <u>Salt:</u>	NaCl	Ant. ½ of sides of tongue		↑↑ Na ⁺ inflow through non-gated Na ⁺ channels
3. <u>Sour:</u>	Acids	Post. ½ of sides of tongue		Closing of acid-sensitive K ⁺ channel → ↓↓ K ⁺ outflow
4. <u>Bitter:</u>	Nicotine & caffeine	Back of tongue	(+) PLC → ↑↑ IP ₃ → ↑↑ Ca ²⁺ release from ER → ↑↑ release of synaptic transmitter	
5. <u>Umami:</u>	Monosodium glutamate	Back of tongue	(?)	

➤ Abnormalities of taste:

1. Ageusia: = absence of taste sense

*Causes:

- (a) Local damage of taste buds: e.g., with glossitis & radiation therapy to head and neck
- (b) Neurological damage of chorda tympani or 9th CN: e.g. by head trauma
- (c) Other causes: e.g., with COVID-19, URTI

2. Hypogeusia: = ↓↓ taste sensitivity

*Causes: SE of bleomycin (chemotherapy)

3. Hypergeusia: = ↑↑ taste sensitivity

*Causes: Autoimmune diseases (as with Addison's disease)

4. Dysgeusia: = distorted taste sense

*Causes: Tongue cancer & SE of cisplatin (chemotherapy)